

10. Cell Phones How young is too young to have a cell phone? A group of eighth-grade boys and girls were surveyed and asked if they had a cell phone, with the following results.

	Cell Phone	No Cell Phone
Boys	52	48
Girls	55	45

- Draw a stacked bar chart to describe the data.
- Draw a side-by-side bar chart to describe the data.
- What can you say about the similarities or differences between boys and girls?

3.01.Q10

HW20240928



DS0309

25. Chirping Crickets Male crickets chirp by rubbing their front wings together. They chirp faster with increasing temperature and slower with decreasing temperatures. The following table shows the number of chirps per second for a cricket, recorded at 10 different temperatures.

Chirps per second	20	16	19	18	18	16	14	17	15	16
Temperature	31	23	33	29	28	24	21	28	21	28

- Which of the two variables (temperature and number of chirps) is the independent variable, and which is the dependent variable?
- Plot the data using a scatterplot. How would you describe the relationship between temperature and number of chirps?
- Find the least-squares line relating the number of chirps to the temperature.
- If a cricket is monitored at a temperature of 27 degrees, what would you predict his number of chirps would be?



DS0311

27. Tall Buildings How closely is the height (in meters) of a tall building and the number of floors related? The heights of 28 buildings in downtown Los Angeles and the number of floors for each building follows.¹⁰

Height	Floors	Height	Floors	Height	Floors	Height	Floors
335	73	220	54	194	49	163	41
310	73	219	52	190	48	162	39
262	62	213	52	189	42	162	40
229	52	213	52	185	45	162	40
228	52	206	49	176	42	158	37
224	55	203	54	174	44	157	40
221	53	197	58	174	44	154	36

Source: World Almanac and Book of Facts, 2017, p. 718.

- Which of the two variables (height and floors) is the independent variable, and which is the dependent variable?
- Find the correlation coefficient, r . What does this tell you about the strength and direction of the relationship between height and number of floors?
- Plot the data using a scatterplot and describe the pattern that you see. Are there any outliers?
- Find the least-squares line, and use it to predict the height of a building in Los Angeles that has 48 floors.

Tree Diagrams Use a tree diagram to find the simple events for the experiments in Exercises 21–24.

21. A bowl contains five candies—red, brown, yellow, blue, and orange. Draw two candies at random, one for you to eat, and one for a friend.

cise 22. A coin is tossed three times and the upper face (head or tail) is recorded for each toss.

23. A coin is tossed four times and the upper face (head or tail) is recorded for each toss.

24. A student was asked four “yes” or “no” questions—do you have an account on Facebook, Twitter, Instagram, and Snapchat.

HW0927

27. A person’s gender and hair color (blonde, brown, black, red, and other) is recorded.

21. Candies A bowl contains five candies—red, brown, yellow, blue, and orange. Draw two candies at random, one for you to eat, and one for a friend. What is the probability that you get the orange candy and your friend does *not* get the red one?

27. Free Throws A particular basketball player hits 70% of her free throws. When she tosses a pair of free throws, the four possible simple events and three of their probabilities are as given in the table:

Second Throw	First Throw	
	Hit	Miss
Hit	.49	.21
Miss	?	.09

- Find the probability that the player will hit on the first throw and miss on the second.
- Find the probability that the player will hit on at least one of the two free throws.

